

C3 Internship Simulation - Project Proposal

Project name	Online Au				
Project Field	☐ Web Development ☒ Java ☐ Robotics ☐ C++ ☐ Cybersecurity ☐ Artificial Intelegence ☐ Game Development ☐ Other				
Number of team members	☐ One ☒ Two ☐ Three ☐ Four ☐ Five				
List of Team Members	Name	Student ID	Grade	Major	GitHub Username
	Alfath Rafiuddin	F25040128	1'st year	Computer Science	Ren-Com
	MOHAMED ELKHAFIF	F24040228	2 nd year	COMPUTE R SCIENCE	med-kha
	Achmad Hilmi Bukhari	F25040127	1'st year	Computer Science	Healme29

1. Project Summary:

The prevalence of fraud nowadays requires special attention. One example is gold scams with misleading claims. Many gold sellers intentionally counterfeit their gold by mixing it with other metals, then sell this alloyed gold at the price of 24-karat gold. Therefore, we present ourselves as a trusted platform for buying and selling gold. Buyers can purchase our gold and store it digitally.

2. Project plan

2.1 Project Objectives: List the main goals of your project.

1. Allow users to create an account (sign-up).
2. Allow users to log in with their existing account (sign-in).
3. Enable users to buy gold using money (IDR).
4. Enable users to buy gold by specifying the weight (grams).
5. Fetch real-time gold price information through an API.
6. Integrate a sandbox payment gateway for deposits and withdrawals.
7. Allow users to deposit money into their account.
8. Allow users to view their current balance.

2.2 Technical Approach: (Explain the technologies, tools, and methods you will use to build your project.)

1. For Application Development:

- ☐ Java
- ☐ NetBeans
- ☐ Swing
- ☐ MySQL

2. For Online Payment:

- ☐ Midtrans Sandbox

3. For Gold Prices:

- ☐ [GoldAPI.io](https://goldapi.io)

2.4 System Design: (Describe the structure or workflow of your system, and include diagrams like flowcharts, block diagrams, system architecture)

This program operates with a simple flow:

When started, the system greets the user and asks if they already have an account. If they do, they must log in by entering their name, username, and password, which are checked against sample (dummy) data. If they don't have an account, they must create a new one by filling in their details and creating a password.

After that, the user enters the main page, which displays a sample balance of Rp20,000,000. They then choose a menu option: buy gold, sell gold, deposit money, or withdraw money.

If they choose to buy, the user can purchase gold in grams or in rupiah. The system calculates the total balance and the amount of gold based on the gold price per gram.

If they choose to sell, the process is reversed: the system reduces the balance according to the amount of gold or money sold, then calculates the remaining gold.

Depositing adds to the balance, while withdrawing reduces it, provided the balance is sufficient.

In every process, the program always checks whether the input is valid (it cannot be negative or exceed the available balance).

After the operation is completed, the system displays the result.

If the user selects "Exit," the program ends.

Flowchart:

https://github.com/Ren-Com/Online_Au/blame/Flowchart/

The flowchart is on github. You need Flowgorithm to see it

2.5 Project schedule: (Outline your weekly plan and what you aim to complete in each stage)

<https://github.com/users/Ren-Com/projects/3/views/1>

2.6 Team Responsibilities: (Specify the tasks and roles of each team member throughout the project.)

1. Alfath Rafiuddin:

- a. develop user authentication (sign-up/sign-in) and session management
- b. handle transactions (buy/sell/deposit/withdraw) and calculate balance/gold accurately
- c. integrate the gold price API (GoldAPI.io)
- d. integrate the payment gateway (Midtrans Sandbox)

2. MOHAMED ELKHAFIF:

- a. build the GUI using Swing/NetBeans
- b. manage program flow, menus, navigation, and input validation
- c. test and debug all features
- d. document the application and user workflow.

Both members will use version control (Git/GitHub), regularly synchronize to integrate backend and frontend, and perform end-to-end testing to ensure all features work properly.

3. Expected Results and Challenges

3.1 Expected Results and Features: (Describe what your project will deliver and the main features it will have)

1. User can make an account
2. User can buy gold by price
3. User can buy gold by weight
4. System can get gold price information through API
5. Sandbox payment gateway can work perfectly
6. User can deposit their money
7. User can see their balance

3.2 Expected Challenges: (List the difficulties you expect to face during the project)

- a. Managing data and state (balance, gold amount, user accounts) so they don't always reset to dummy values.
- b. Handling Scanner input issues, especially when mixing `nextInt()` and `nextLine()`.
- c. Designing a clean and modular program structure, since the current functions are mixed and deeply nested.
- d. Performing accurate financial calculations, including dealing with floating-point precision.
- e. Building a real login and sign-up system, which requires actual data storage or a database.
- f. Fetching gold prices through an API, which involves making HTTP requests and parsing JSON.
- g. Integrating a payment gateway, including authentication and handling responses.
- h. Implementing error handling and input validation so the program doesn't crash.